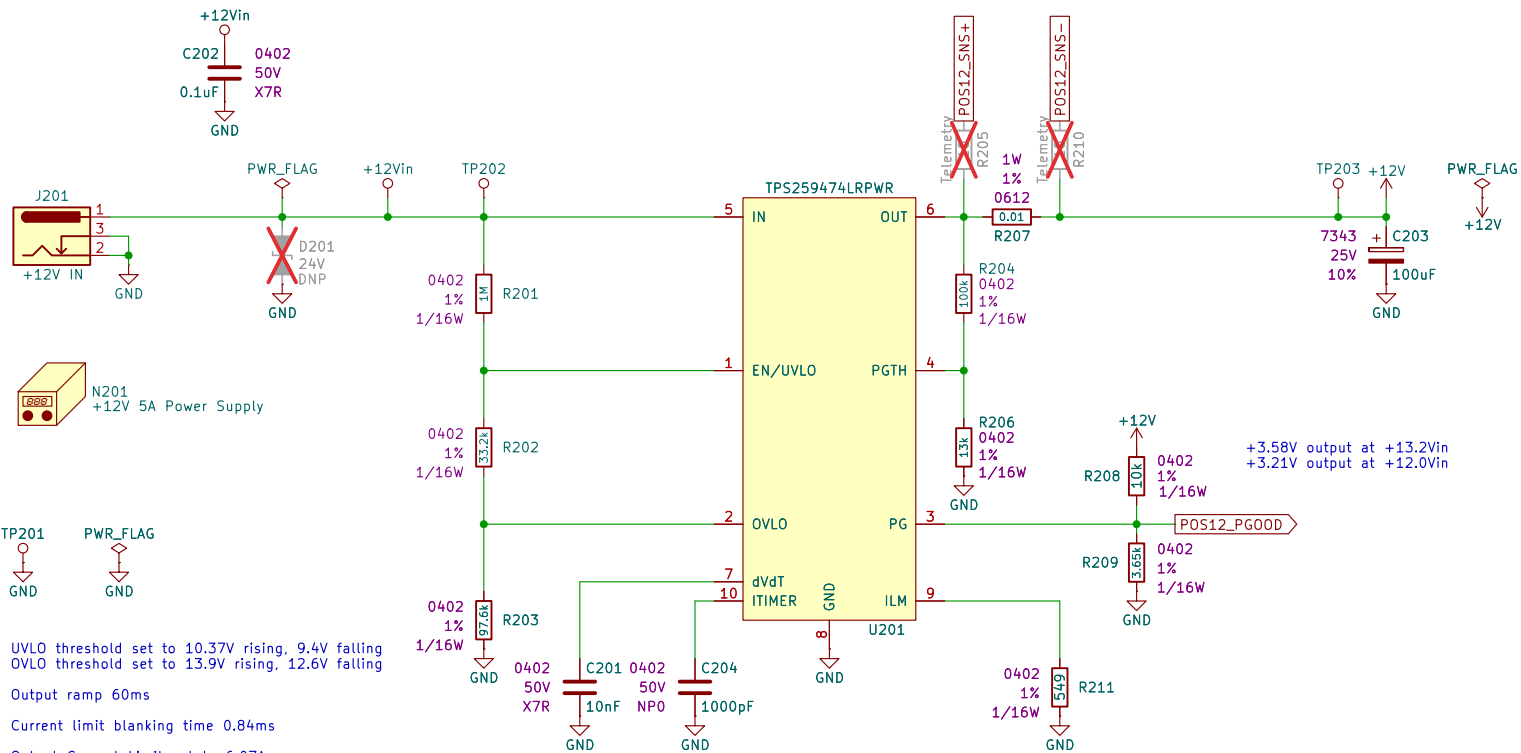


1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch	Platform ETC File: platform_etc.kicad_sch			A
	12V Telemetry File: POS12_telemetry.kicad_sch	PGOOD LEDs File: pgood_leds.kicad_sch			
	3.3V Power Supply File: POS3P3_power_supply.kicad_sch	Status LEDs File: status_leds.kicad_sch			
	3.3V Telemetry File: POS3P3_telemetry.kicad_sch	RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch			
B	dsPIC33 Programming File: dspic33_programming.kicad_sch	Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch			B
	dsPIC33 Power File: dspic33_power.kicad_sch	Source Selection, PGAs File: source_selection_pgas.kicad_sch			
	dsPIC33 GPIO File: dspic33_gpio.kicad_sch	Anti-alias Filtering File: anti_alias_filtering.kicad_sch			
	USB UART Bridge File: usb_uart_bridge.kicad_sch	Audio Voltage References File: audio_voltage_references.kicad_sch			
		LED Panel Buffers File: led_panel_buffers.kicad_sch			
C		Pushbuttons File: pushbuttons.kicad_sch			C
		Mechanical File: mechanical.kicad_sch			
		LED Panel Connectors File: led_panel_connectors.kicad_sch			
D					D
1	2	3	4	5	



Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A @ 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

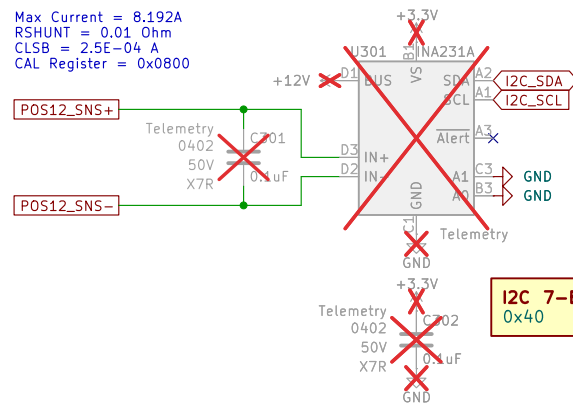
Sheet: /12V Power Input/
 File: POS12_power_input.kicad_sch

Title:

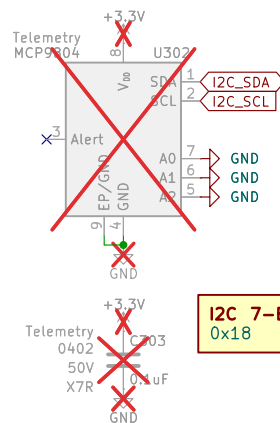
Size: A Date: 2026-10-08
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Rev: A
 Id: 2/21

Max Current = 8.192A
 RSHUNT = 0.01 Ohm
 CLSB = 2.5E-04 A
 CAL Register = 0x0800



I2C 7-Bit Address:
 0x40



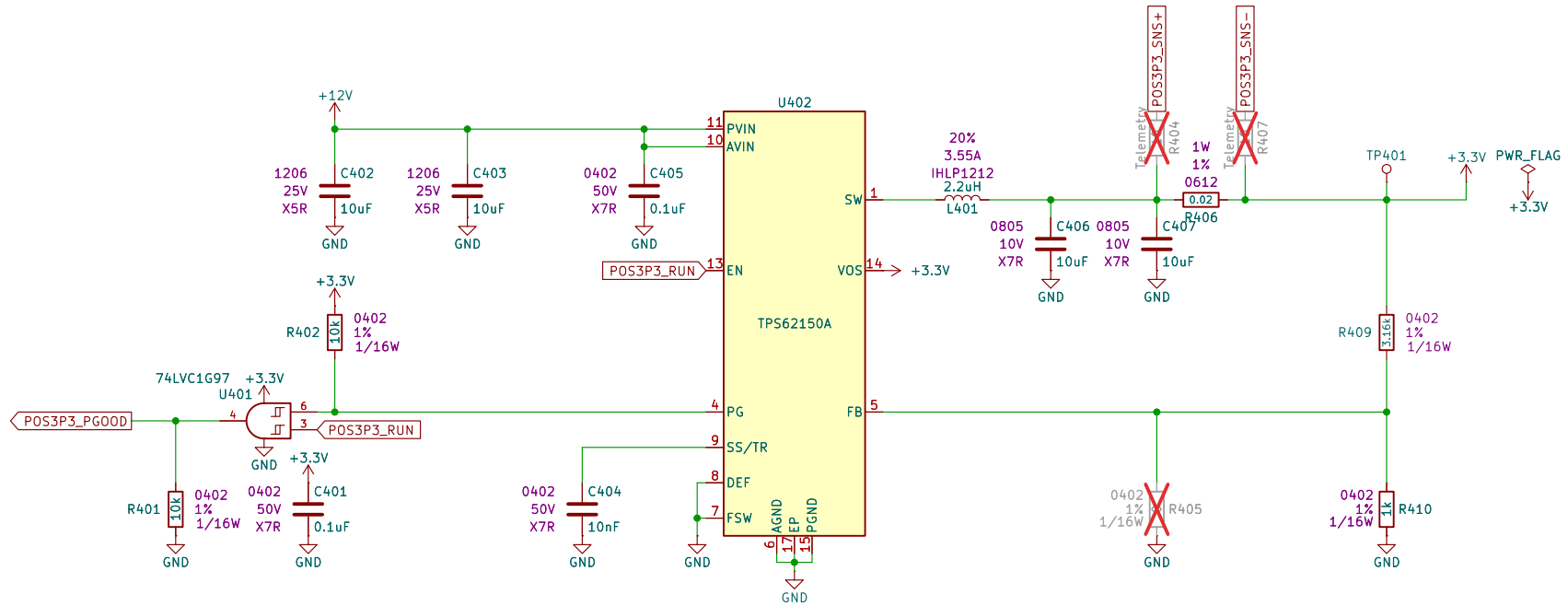
I2C 7-Bit Address:
 0x18

Sheet: /12V Telemetry/
 File: POS12_telemetry.kicad_sch

Title:

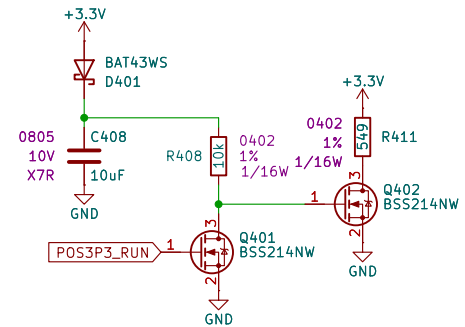
Size: A Date: 2026-10-08
 KiCad E.D.A. 10.0.3

Rev: A
 Id: 3/21



Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog_Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V_Reference	0.2mA	1	0.2mA
PCA9536_GPIO_Expander	0.29mA	1	0.29mA
I2C_Buffers	8.3mA	3	24.9mA
Octal_Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary_Encoder_Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap_Touch_Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA

POS12_PG00D 0 POS3P3_RUN
R403

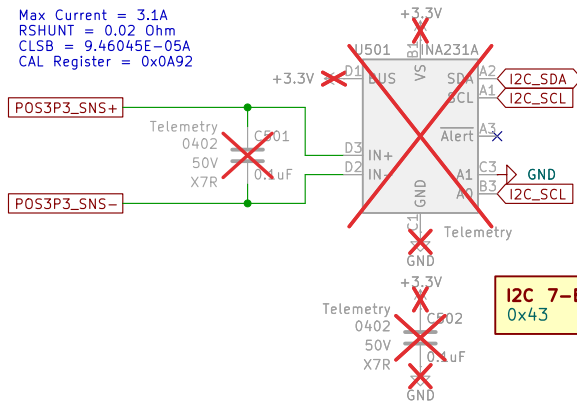


Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

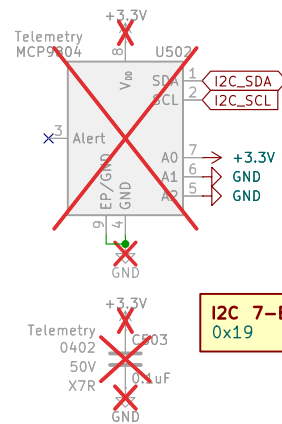
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Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 4/21



I2C 7-Bit Address:
 0x43



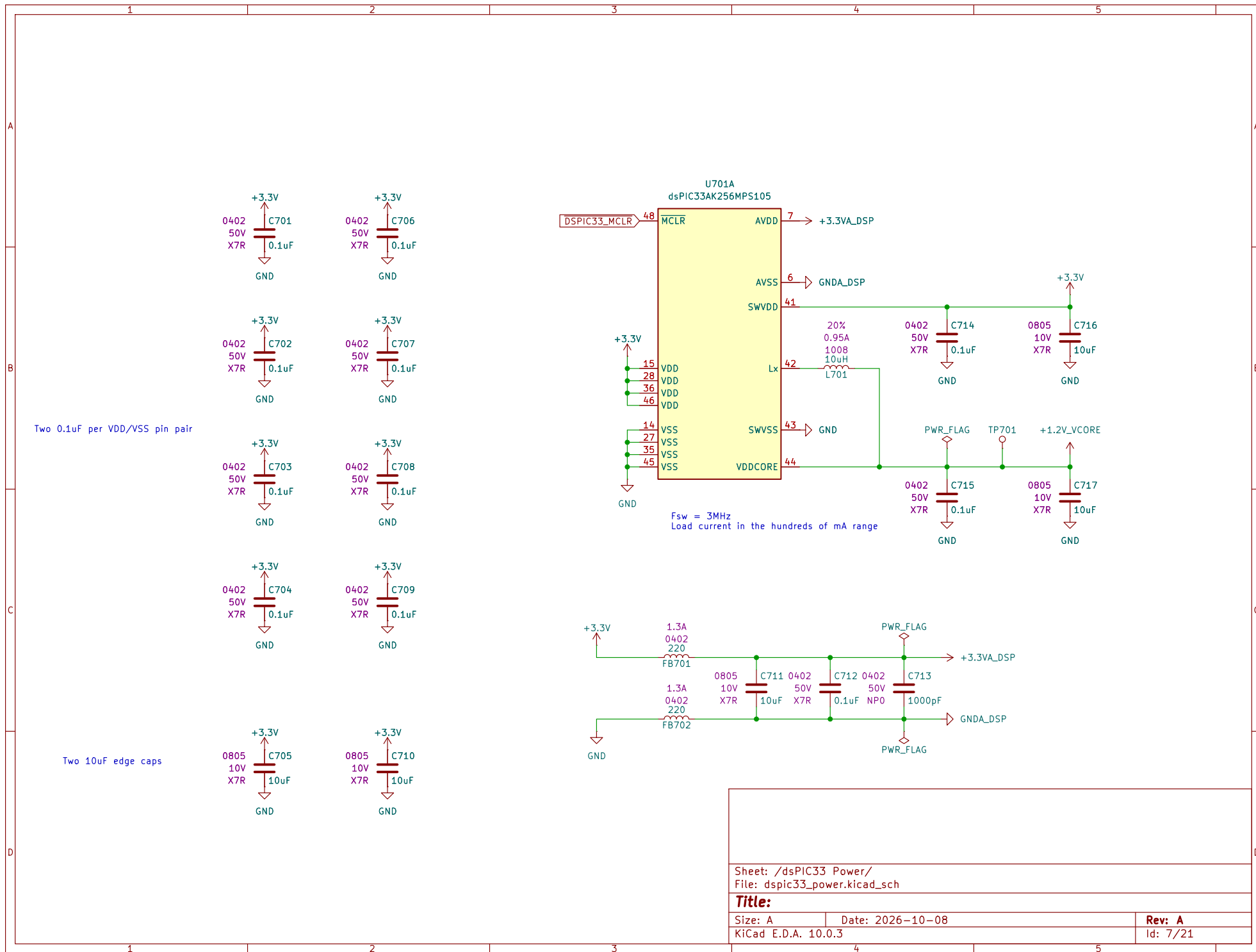
I2C 7-Bit Address:
 0x19

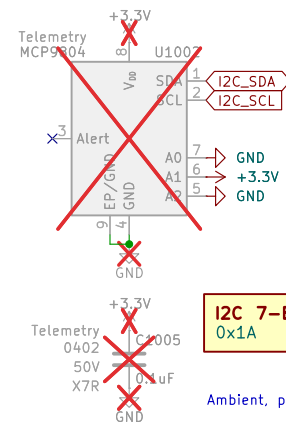
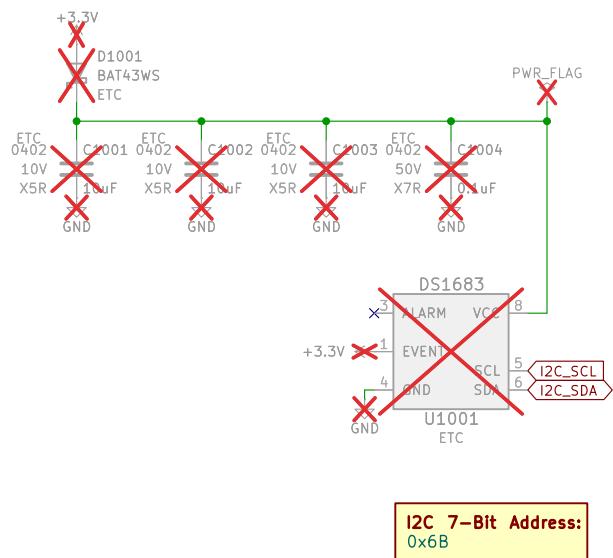
Sheet: /3.3V Telemetry/
 File: POS3P3_telemetry.kicad_sch

Title:

Size: A Date: 2026-10-08
 KiCad E.D.A. 10.0.3

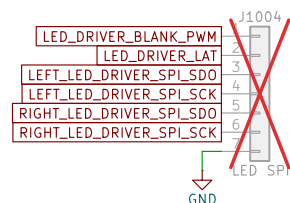
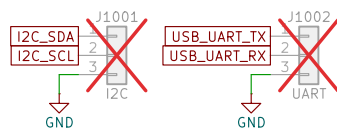
Rev: A
 Id: 5/21





Ambient, place away from power components

Test Points

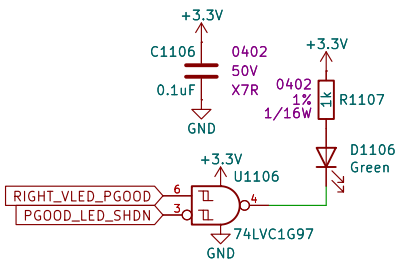


Sheet: /Platform ETC/
File: platform_etc.kicad_sch

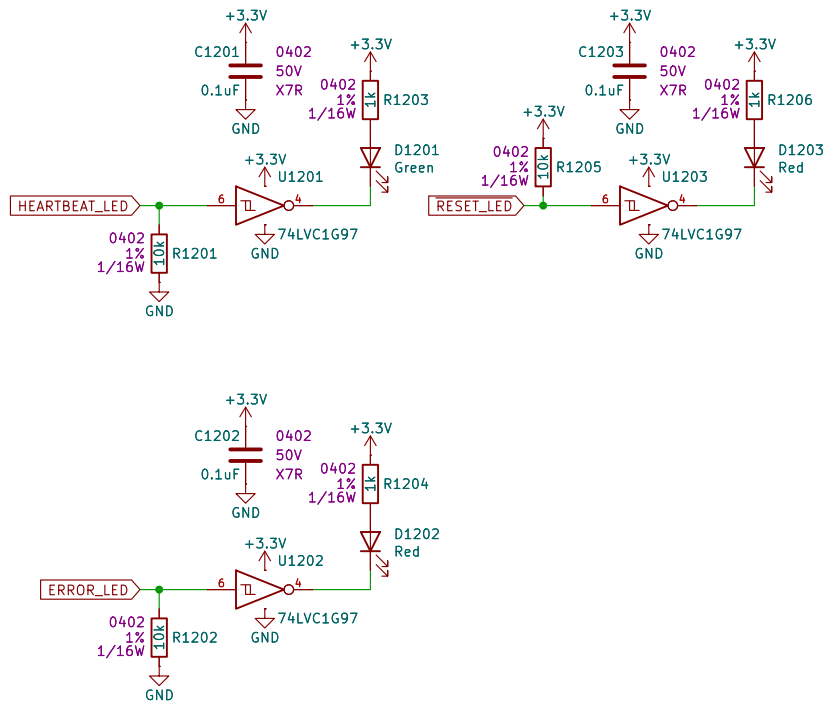
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KiCad E.D.A. 10.0.3

Rev: A
Id: 10/21



Rev: A
Id: 11/21

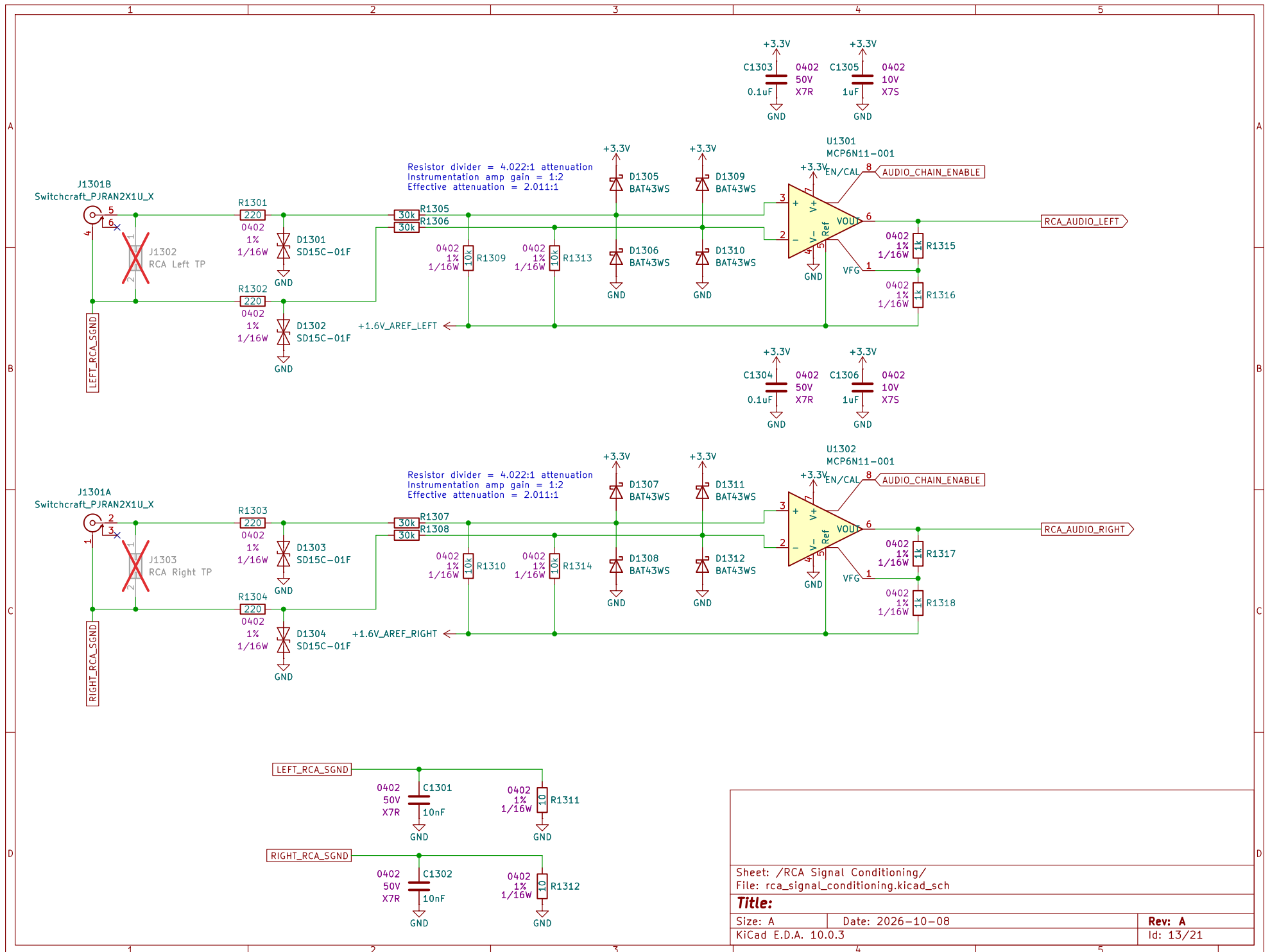


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 12/21



Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

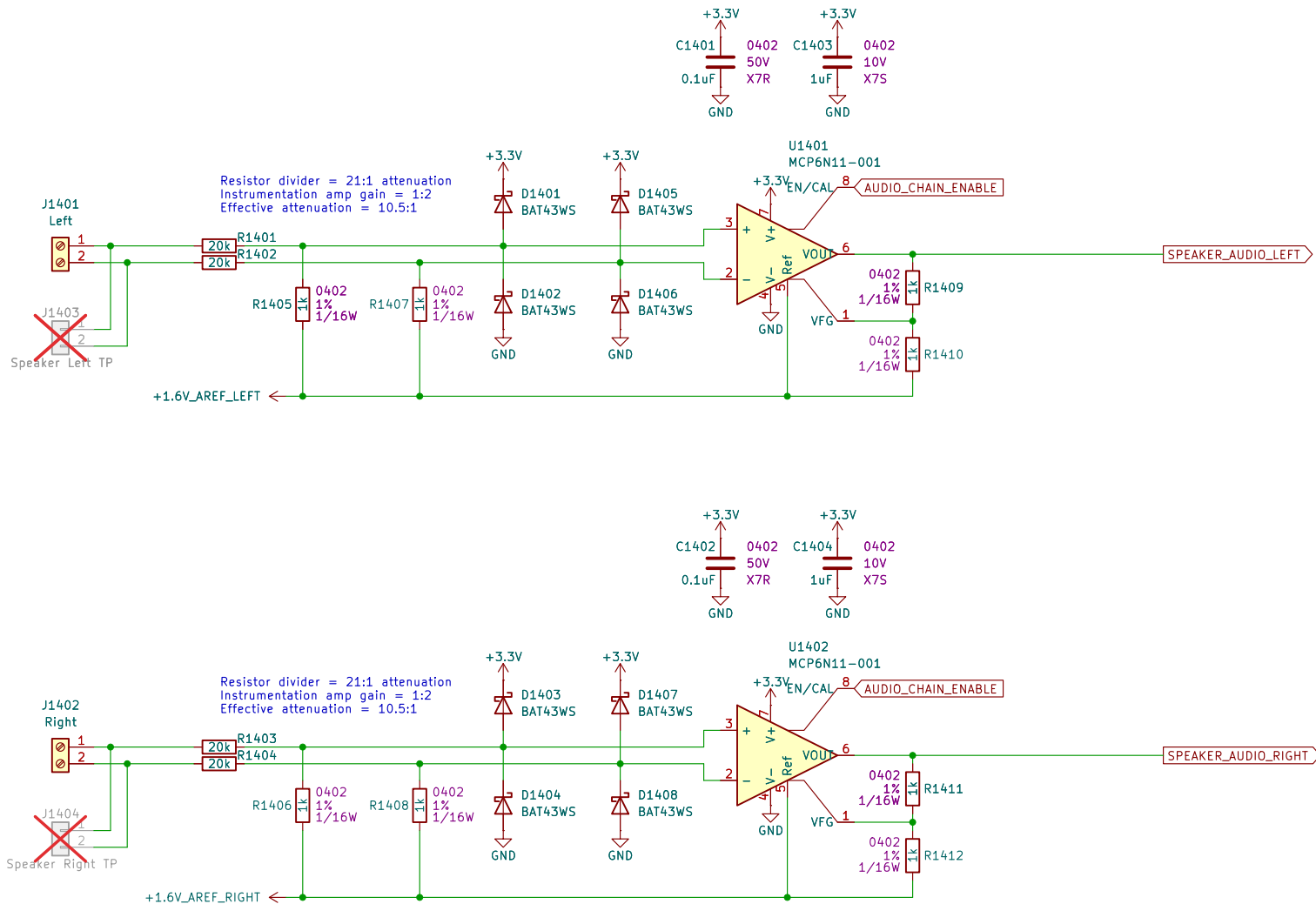
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Size: A Date: 2026-10-08

KiCad E.D.A. 10.0.3

Rev: A

Id: 13/21

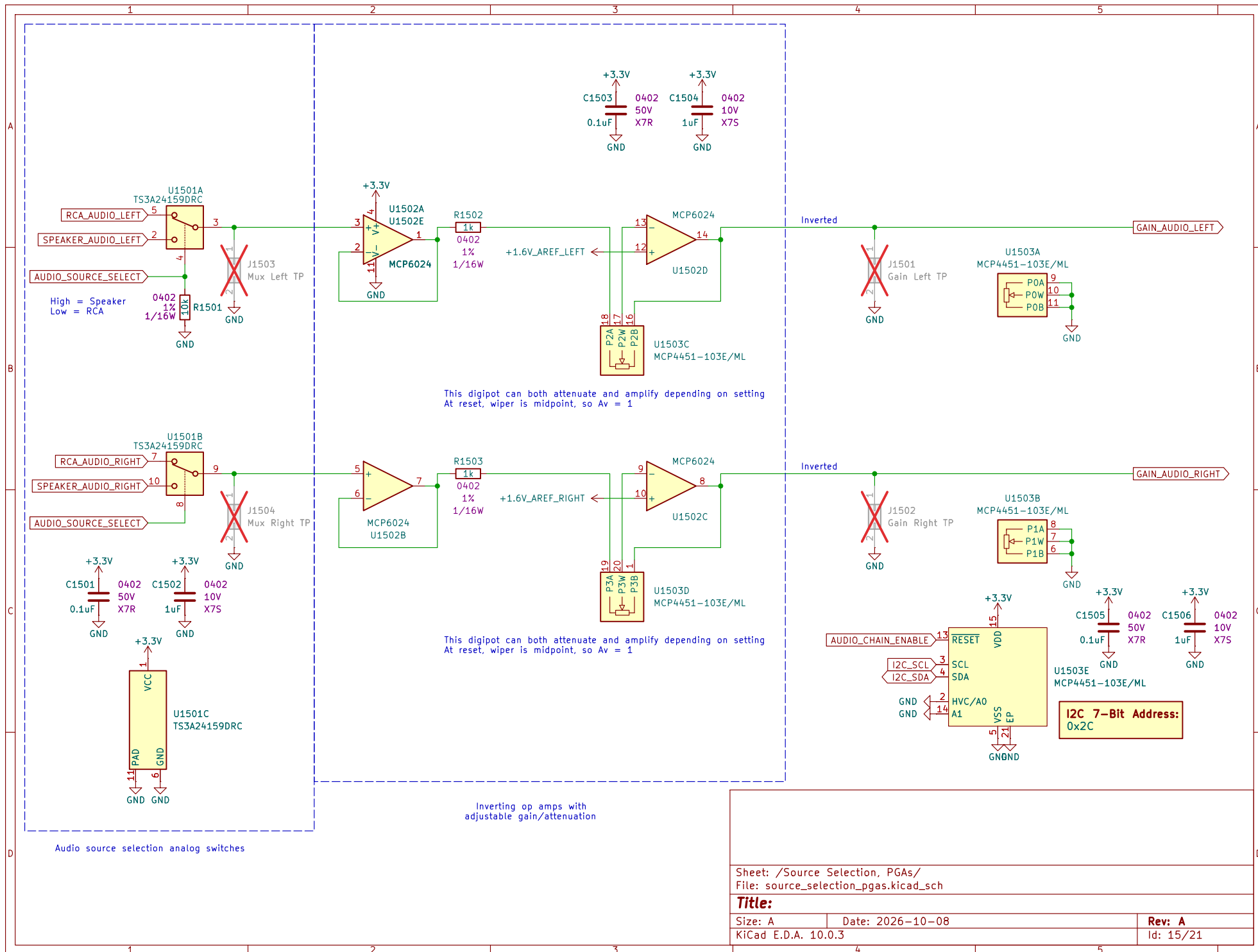


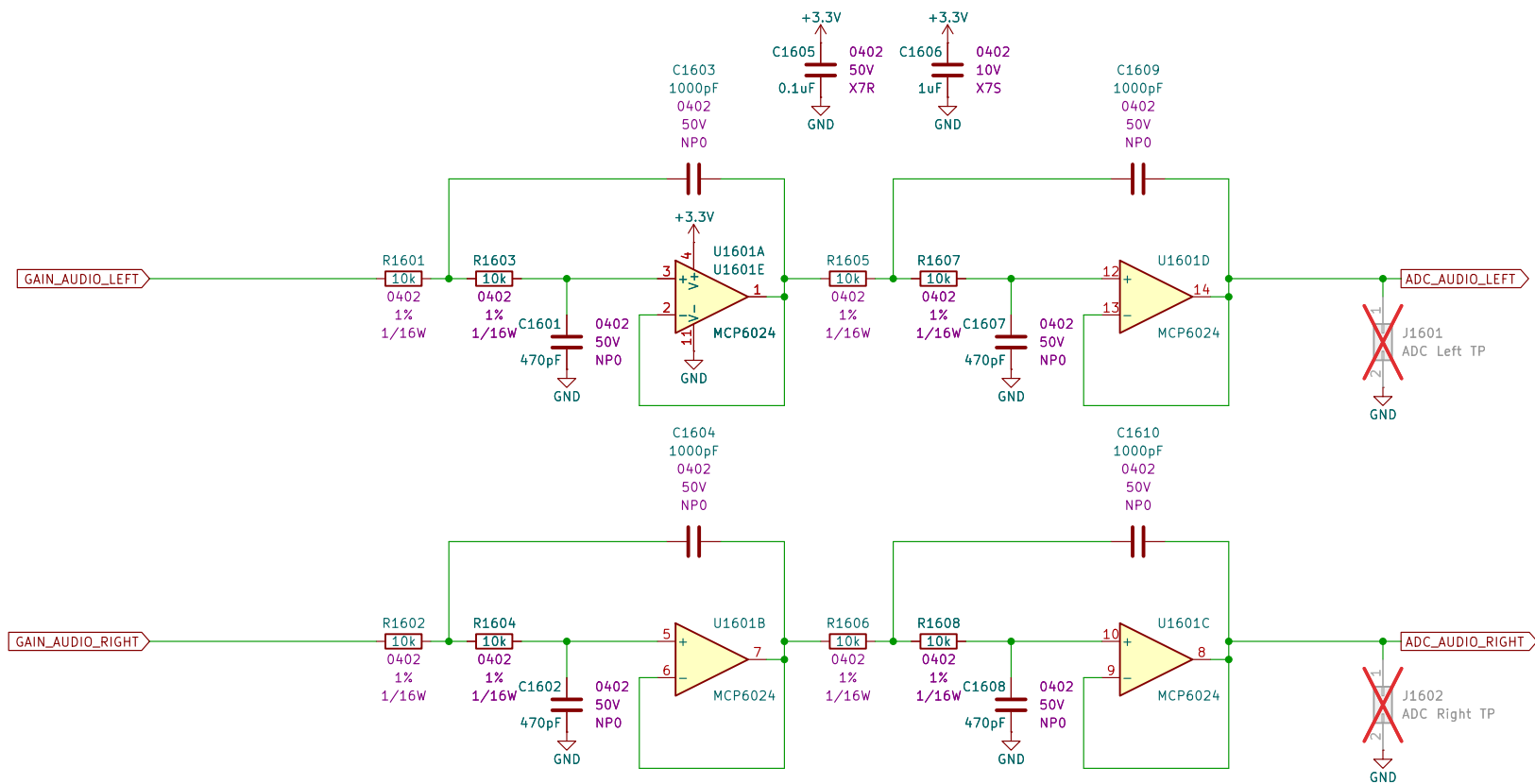
Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 14/21





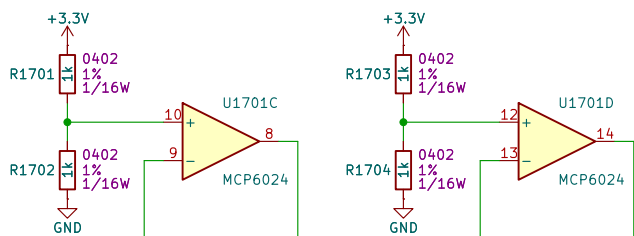
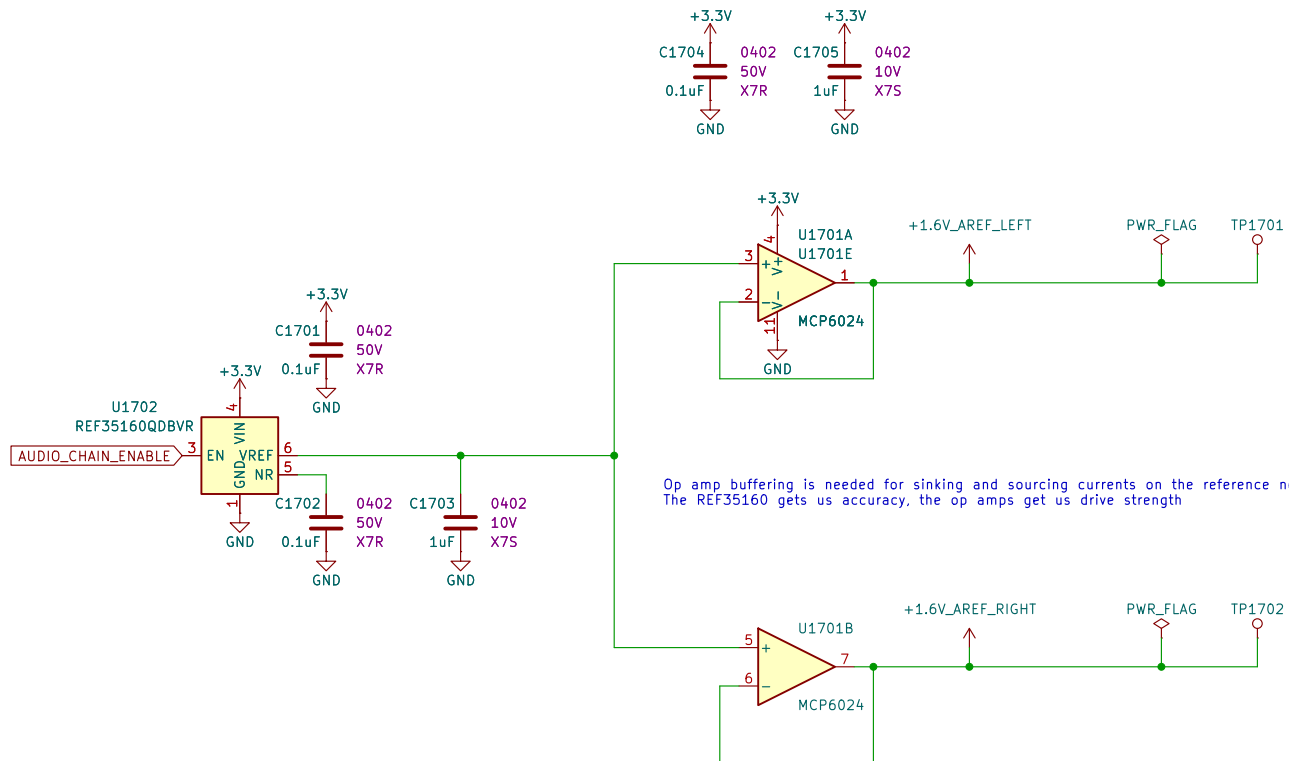
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard around this frequency

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2026-10-08
 KiCad E.D.A. 10.0.3

Rev: A
 Id: 16/21

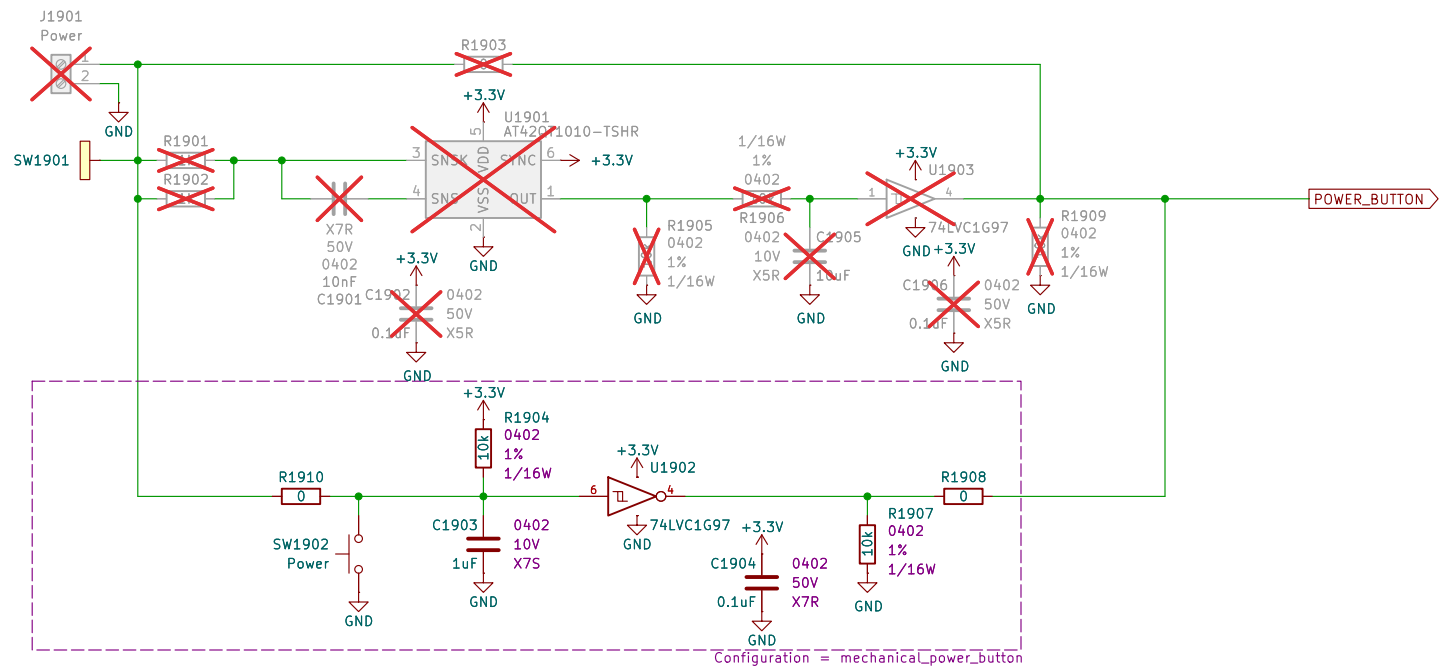


Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

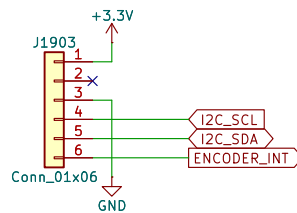
Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 17/21



N1901
Adafruit Rotary Encoder Module

I2C 7-Bit Address:
0x36

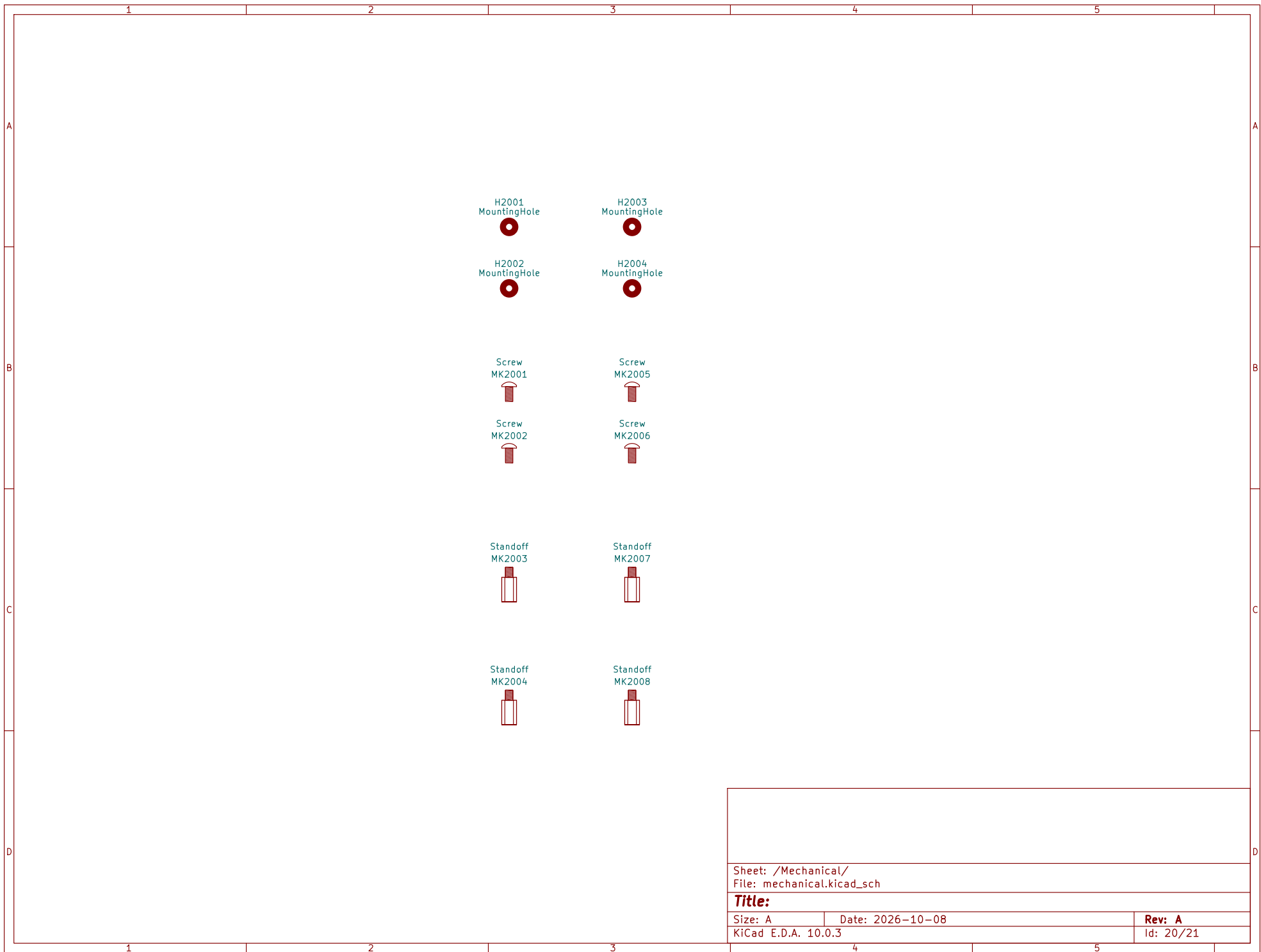


Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A Date: 2026-10-08
KiCad E.D.A. 10.0.3

Rev: A
Id: 19/21



Sheet: /Mechanical/ File: mechanical.kicad_sch		
Title:		
Size: A	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.3		Id: 20/21

